

Hypertensive Disorders of Pregnancy

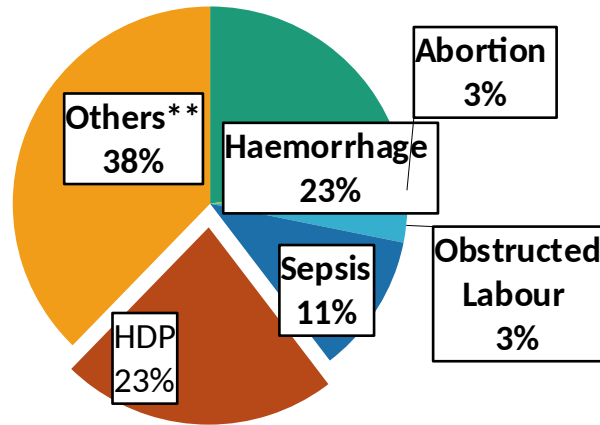
Topic 1



Importance of diagnosing hypertensive disorders of pregnancy

Hypertensive disorders are the second leading cause of maternal mortality in India

Causes of Maternal Death MDR 25-26 (Odisha)



- Hypertensive disorders increase the **risk of complications** in pregnant women, her foetus and new born
- Accurate measurement and recording of pregnant mother's BP and urine examination helps in **timely diagnosis** of at risk mothers
- **Early diagnosis and timely management and/or referral** helps in reducing the risk of complications in such cases
- During pregnancy **Calcium supplements** can help in prevention of hypertensive disorder during pregnancy

Effects of high BP on mother and child during pregnancy

Hypertensive disorders increase the chances of death in both mother, fetus and newborn. Early diagnosis and referral helps in reducing this risk.

Effects on mother:

- ☐ Antepartum Haemorrhage
- ☐ Postpartum Haemorrhage
- ☐ Preterm labour
- ☐ Maternal Death

Effects on foetus and Newborn

- ☐ Intrauterine Growth Restriction
- ☐ Intrauterine Death
- ☐ Prematurity
- ☐ Low birth weight

Diagnostic Criteria

Before 20 weeks of

Chronic Hypertension

Not induced by pregnancy

After 20 weeks of Pregnancy

Pregnancy Induced Hypertension

Gestational Hypertension

Increased BP and protein in urine
BP more than or equal to 140/90

Pre-Eclampsia

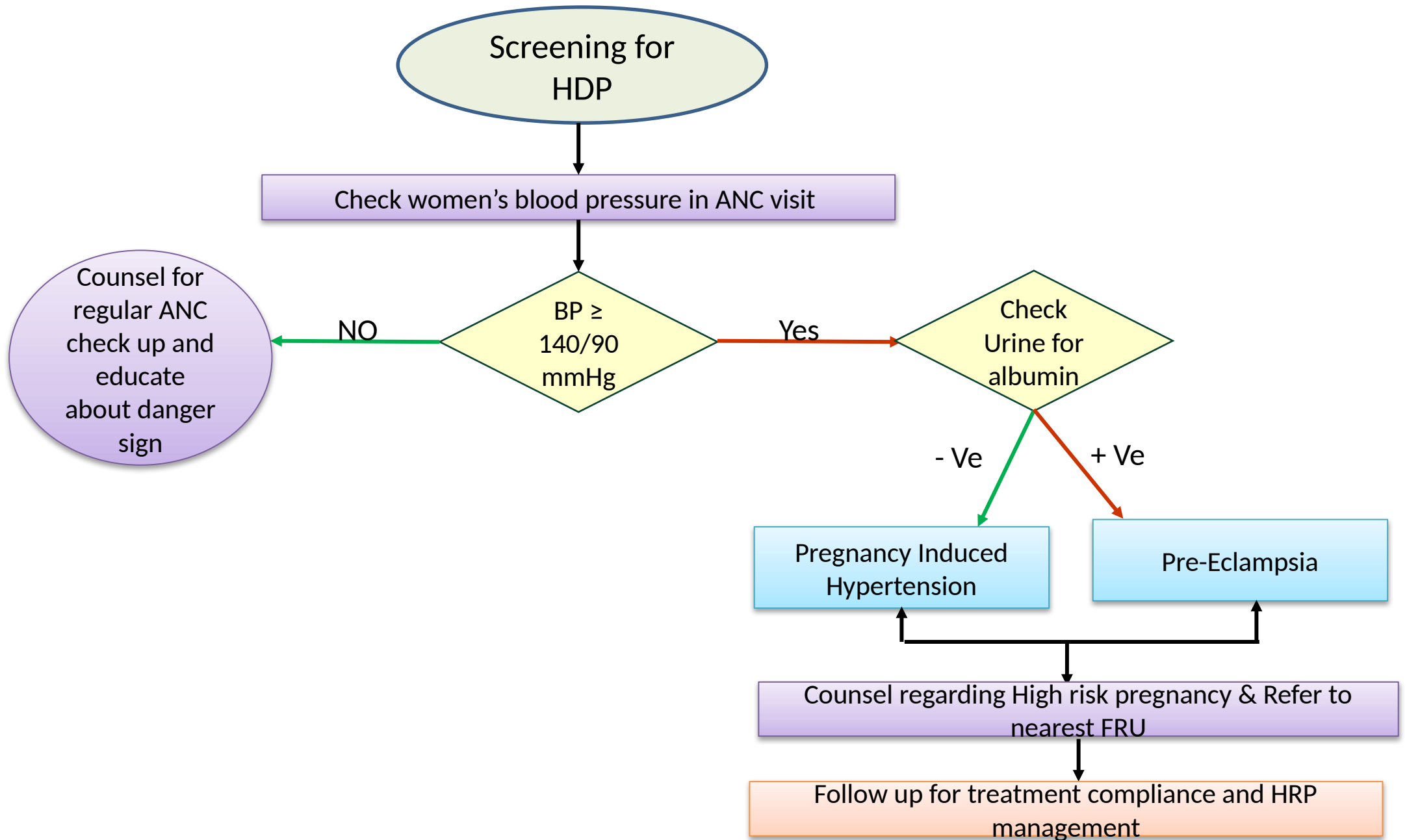
Increased BP and protein in urine
BP more than or equal to 140/90 and less than 160/110
And Proteinuria Trace/1+/2+

Severe Pre-Eclampsia

Increased BP and protein in urine
BP more than and equal to 160/110
Heavy Proteinuria 3+/4+
And /or Presence of danger signs
Pre Eclampsia + Danger signs

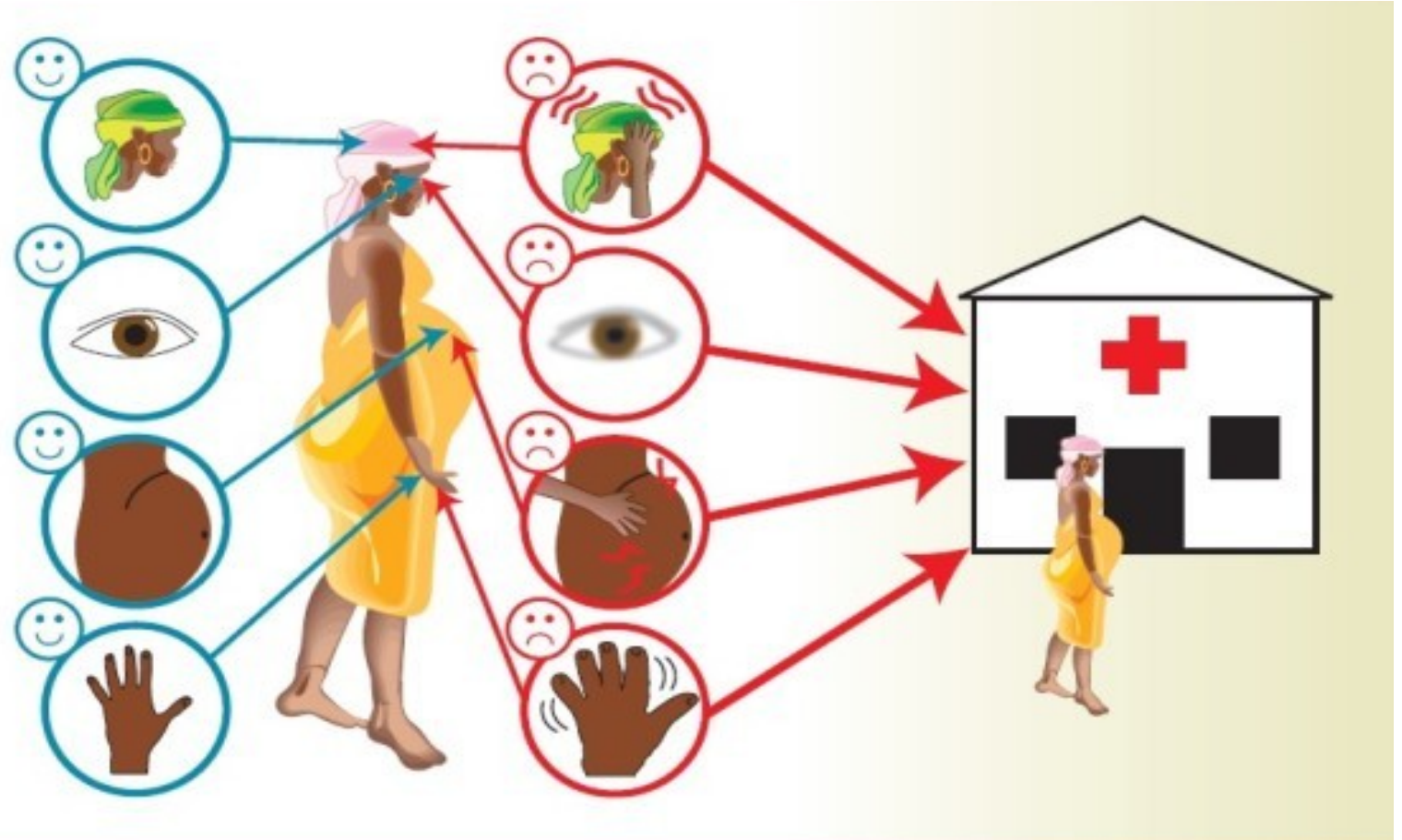
Eclampsia

Increased BP, protein in urine and convulsions
BP more than and equal to 140/90
Proteinuria Trace or more
Presence of convulsions
Convulsions in pregnancy, labor and postpartum period should be considered 'Eclampsia' unless proved otherwise.



Danger Signs during Severe Pre-Eclampsia and Eclampsia

- Headache (increasing in frequency & severity)
- Blurring of vision
- Oliguria (less than 500 ml in 24 hours)
- Pain in upper part of abdomen
- Abnormal oedema over face, hands, abdomen and vulva



Need for MgSO₄

Management with Inj. MgSO₄ should be given in following conditions:

- Eclampsia :
 - ✓ Convulsions
- Severe PE:
 - ✓ BP ≥ 160/110 with proteinuria 3+ or 4+
 - ✓ PE with presence of danger signs like headache, blurring of vision, epigastric pain or oliguria and abnormal edema over face, hands, abdomen and vulva

Administration of MgSO₄

- **First dose (at SC/PHC level (If No MO)/referral dose: Total 10 grams**
 - ✓ 5 g (10ml) magnesium sulphate deep IM in each buttock
 - ✓ Patient should reach FRU in 2 hours for further management.



As per GoI guidelines, the pre-referral dose can be given by staff nurse/ANM/CHO, before referring the woman to a higher center, record the same in MCP card.

Need for MgSo4

Administration of Mgso4 at Facility

- **Loading dose: Total 14 grams**
- ✓ 4 g (8mL) magnesium sulphate diluted with 12 ml NS or distilled water in 20 ml syringe i.e. 20%, and given slow IV in 5-10 minutes
- ✓ 5 g (10mL) magnesium sulphate with 1 ml 2% lignocaine deep IM in each buttock

Maintenance dose

- ✓ 5 g (10mL) magnesium sulphate with 1 ml 2% lignocaine deep IM in alternate buttock every 4 hours
- ✓ To be given till 24 hours after last convulsion or delivery - whichever occurs later

Dose of Recurrent convulsions

- ✓ If 30 minutes after loading dose – repeat MgSO4 2 gm 20% (4ml drug in 6ml NS) slow IV in 5 minutes (Can be given only once).

MgSO₄ TOXICITY SIGNS



Watch for toxicity before every maintenance dose if



Deep tendon reflex
(Knee jerk):

Absent



Urine output:

<25–30 ml/hr



Respiratory rate:

<16 breaths/min



NOTE:

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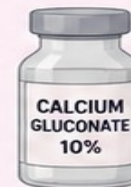
If urine output is less or knee reflex is absent, withhold the next dose of MgSO₄ and monitor hourly and restart when signs of toxicity are absent.



2



If respiratory rate is less: < 12/min
Give antidote Inj. Calcium gluconate 1gm
(10 ml 10 % solution in 10 minutes)



Do not give any sedatives or tranquilizers



Role of Anti-hypertensive



- **Tab Labetalol** to be used for controlling BP .
Dosage 100mg 8-12 hourly (PIH/ Preeclampsia)

In case of Severe Preeclampsia/ Eclampsia , Use of



• Medication:

- **Tab Nifedipine** stat 10 mg orally Immediate release, after 20 mins if BP Not control repeat 10-20 mg after 20 mins if BP not control repeat 10-20 mg (Max 30 mg) : (DON'T use capsule)



Note: If women is uncouncious tablet can be given through ryle's tube.



- **Inj. labetalol** is recommended for initial control of BP, 20 mg IV bolus, repeat 40 mg after 10 mins, if Bp not control repeat 80mg every 10 mins (Max 300 mg), with cardiac monitoring



- Hospitalize, reassure, advice rest



- Start anti-hypertensive agent when SBP $\geq$ 140mm Hg and or diastolic $\geq$ 90mm Hg



- Investigate — CBC with peripheral smear and platelet count, LFT, KFT and fundus exam



- BP and urine output monitoring